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Part 1

Contract Monthly Claim System (CMCS) Prototype Report

Database Design

The database (Figure 1) has been designed to facilitate access control and accountability by using user roles and claim approval logging.

Lecturers have been linked to specific modules and modules have a *scheduled sessions* field to facilitate automated claim validation. This structure improves the user experience; module options on the claim form are limit to those a lecturer teaches, and those performing approvals know that a lecturer can't claim for a module they don't teach.

Supporting document and invoice tables are used to associate a claim with documents stored in external systems.

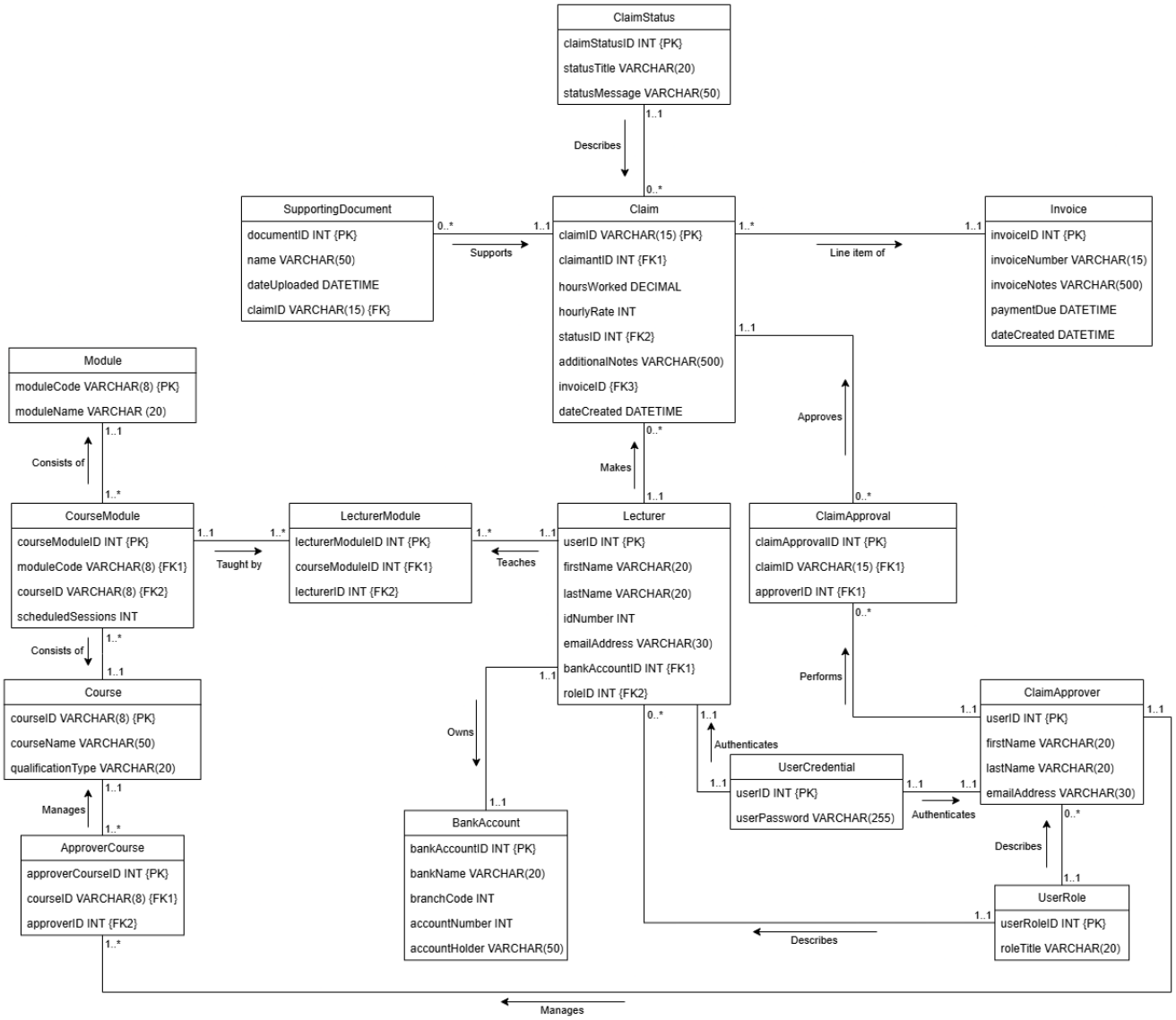


Figure 1 - CMCS UML class diagram for databases.

Graphical User Interface (GUI)

Theme

For the CMCS, I selected colours and font found on the VC website. Using VC brand colours provides a familiar feel for the application.

CMCS buttons are presented with a drop shadow to make them appear elevated, communicating clickability.

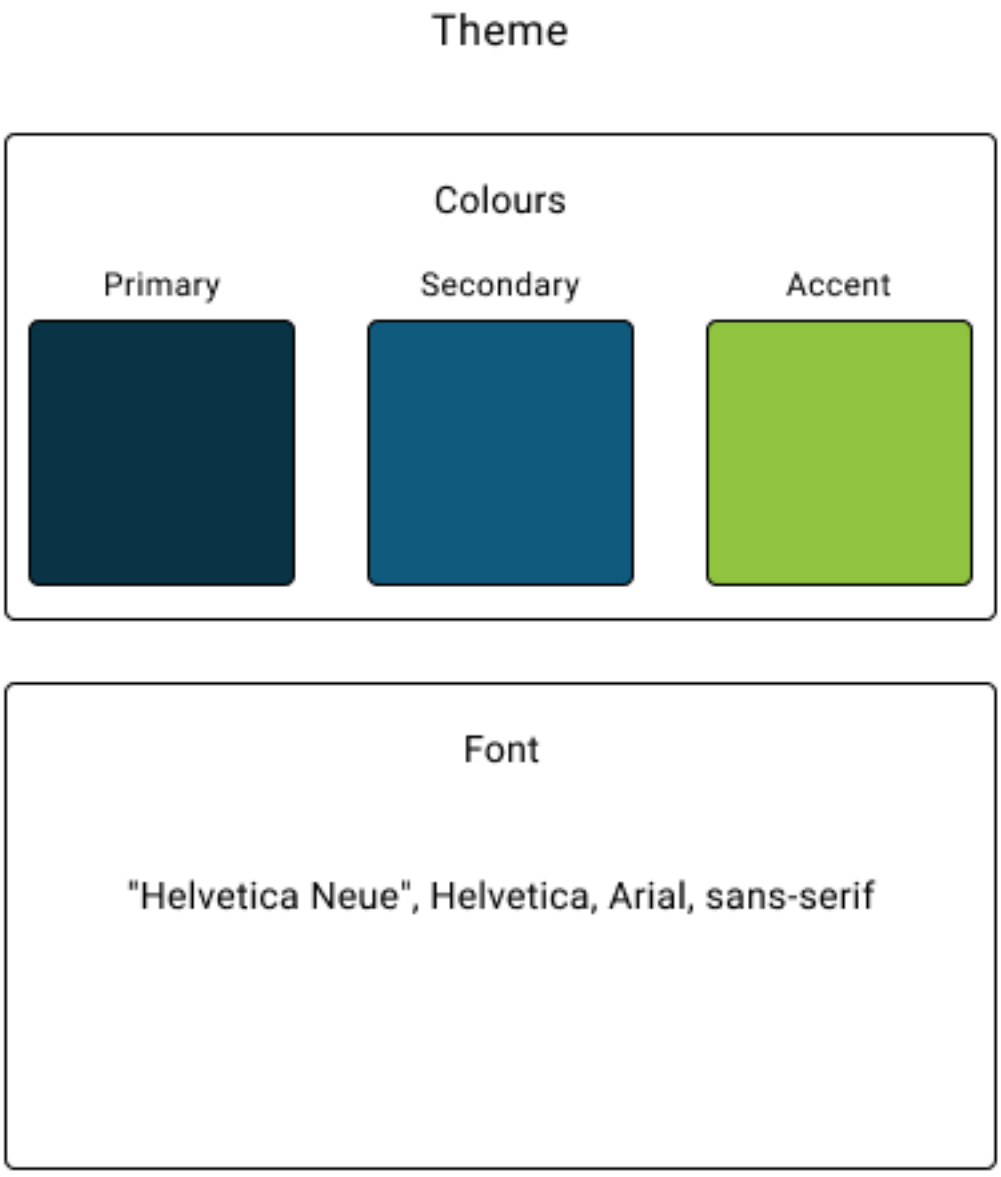


Figure 2 - Colour palette and Font for the CMCS.

Login

All users are presented with the same simple login interface. The interface flow is vertical, providing the visitor with required information in a logical sequence. All controls are clear in their purpose and the sign in procedure is intuitive. After signing in, there is a simple header on all views containing a 'Sign Out' button, an account avatar icon which can be used to navigate to the user's account view, and if applicable, a 'Back' button.

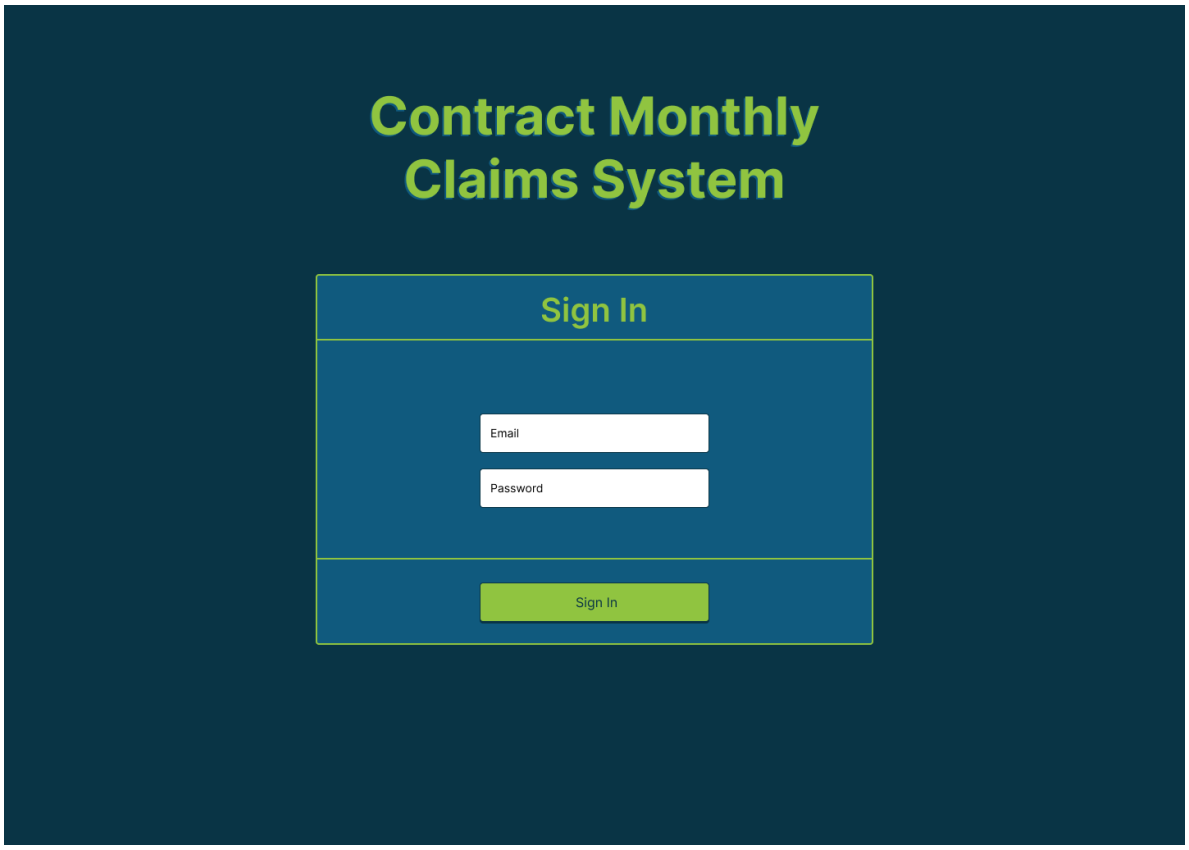


Figure 3 - the login interface for all users.

Programme Coordinator (PC) & Academic Manager (AM)

After signing in, by clicking 'Sign In' (Figure 3), PCs and AMs will be met with a dashboard (Figure 4) displaying claims created by lecturers. Claims are filtered using the vertical menu left of the table. The active filter is highlighted.

Each claim has a status with a corresponding colour. Yellow grabs attention, implying needed action, green indicates that a claim is approved, and black indicated that the claim is terminated.

Essential information is shown for each claim. In the ‘Actions’ column, the user can choose to view more detailed information for the claim, or to approve or reject the claim based on the basic information.

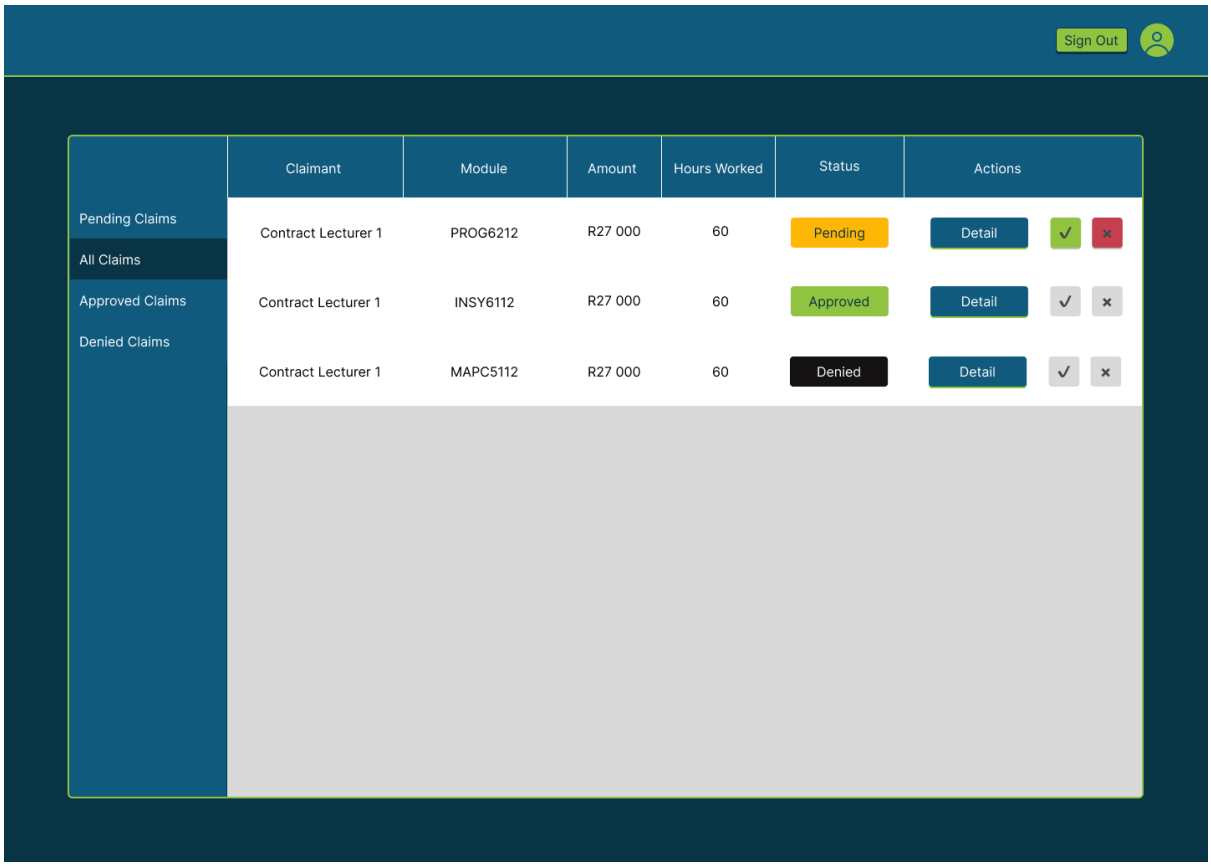


Figure 4 - Programme Coordinator and Academic Manager home view.

Figure 5 shows an example of the modal which users will be presented with if they click the ‘Details’ button (Figure 4). The modal shows the user whether another user has already approved the claim or not (each claim requires approval from a PC and AM).

Figure 6 shows an example of the confirmation dialogue show to users when they click the checkmark button on a claim. This behaviour will exist for the reject (cross/x) button to prevent accidental rejection.

This design allows users to scrutinise and approve claims with a minimal number of clicks, and presents information to the user in a logical order.

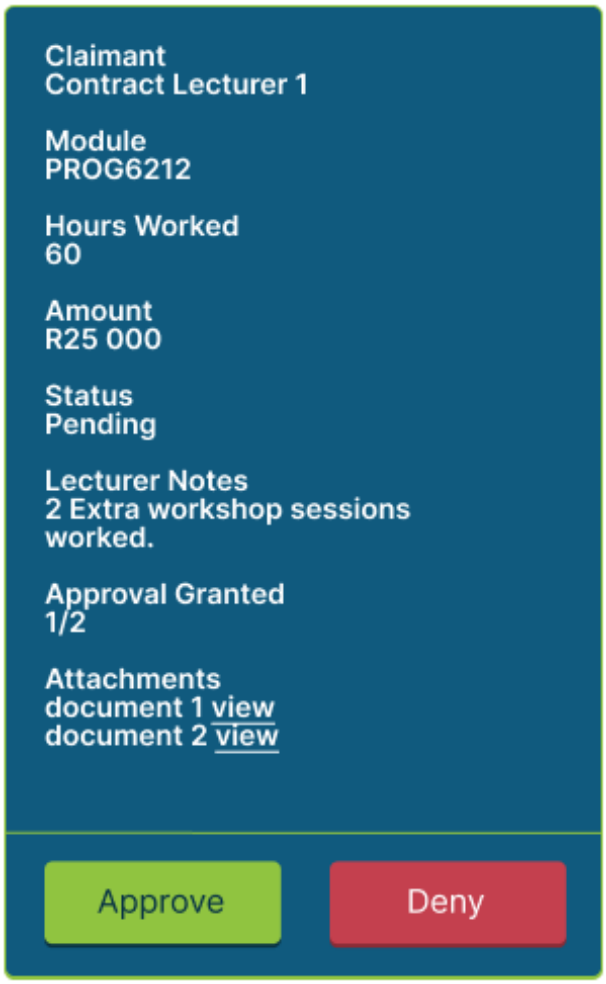


Figure 5 - Claim details modal.

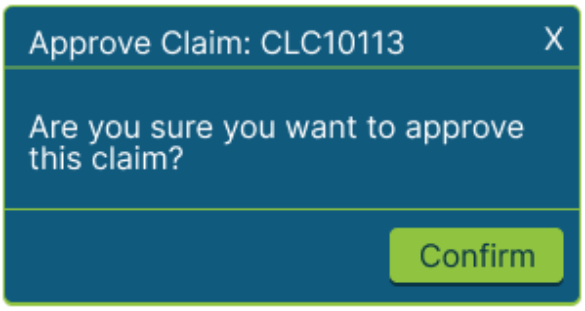


Figure 6 - Confirmation pop-up for claim approval.

Contractor Lecturer (CL)

After successfully signing in, CLs will be presented with the view shown in Figure 7. Navigation choices are clearly communicated; make a new claim or view their claim history.

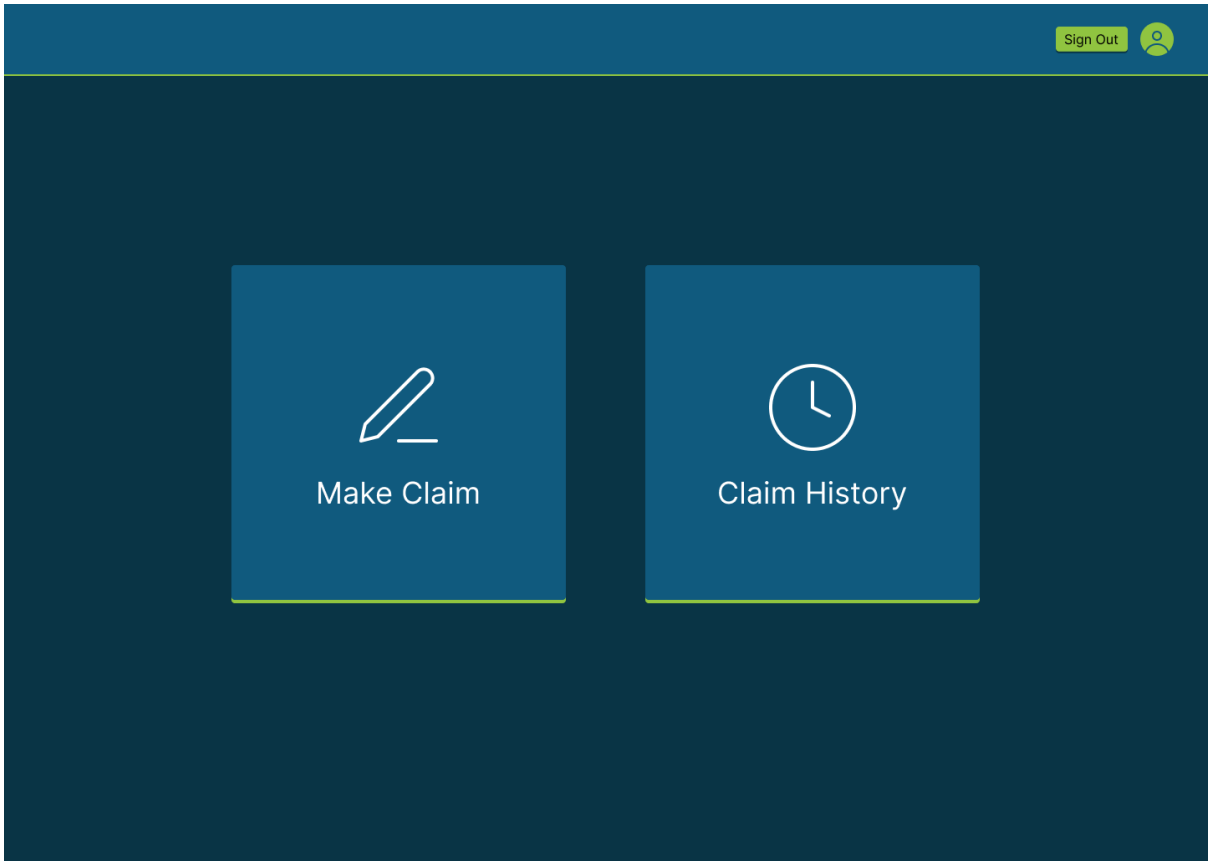


Figure 7 - Contractor lecturer home

'Make Claim' will show the view seen in Figure 8. View information is communicated to the user sequentially. If the user navigated in error, they can click the 'Back' button.

The 'New Claim' heading orientates the user.

The claim form prompts the user to select the module they wish to make a claim for from a select input (required), then enter the hours worked (required), their hourly rate (required), add additional information they think important (optional), and attach supporting documentation (optional) with the 'Add Attachment' button. An accented button is presented for the user to submit their claim.

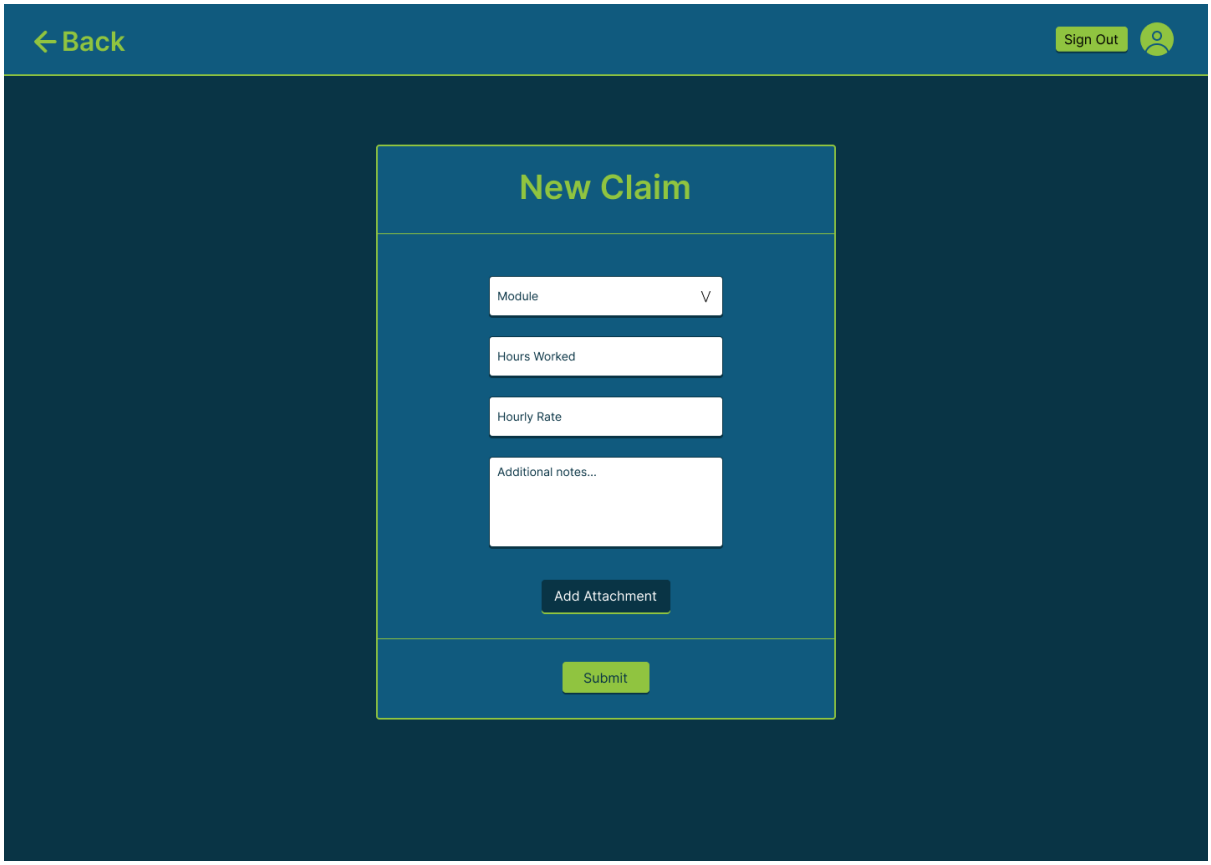


Figure 8 - Claim creation view.

'Claim History' (Figure 7) navigates to the view is seen in Figure 9. Here the lecturers can see the same information about the claims as the PCs and AMs for process transparency.

If a claim was made in error, or the lecturer made a mistake, claims can be withdrawn using the 'Delete' button.

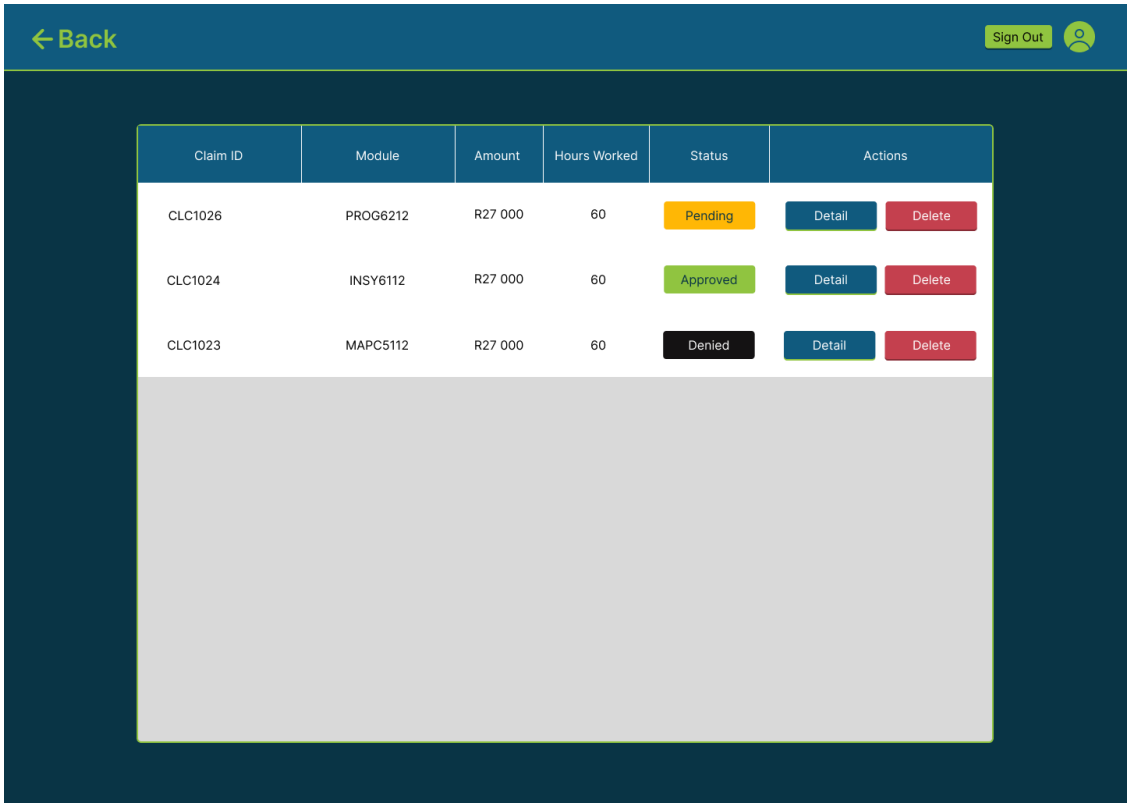


Figure 9 - Claim history view.

Figure 10 shows the user account view. Lecturers can update their personal information and banking details here.

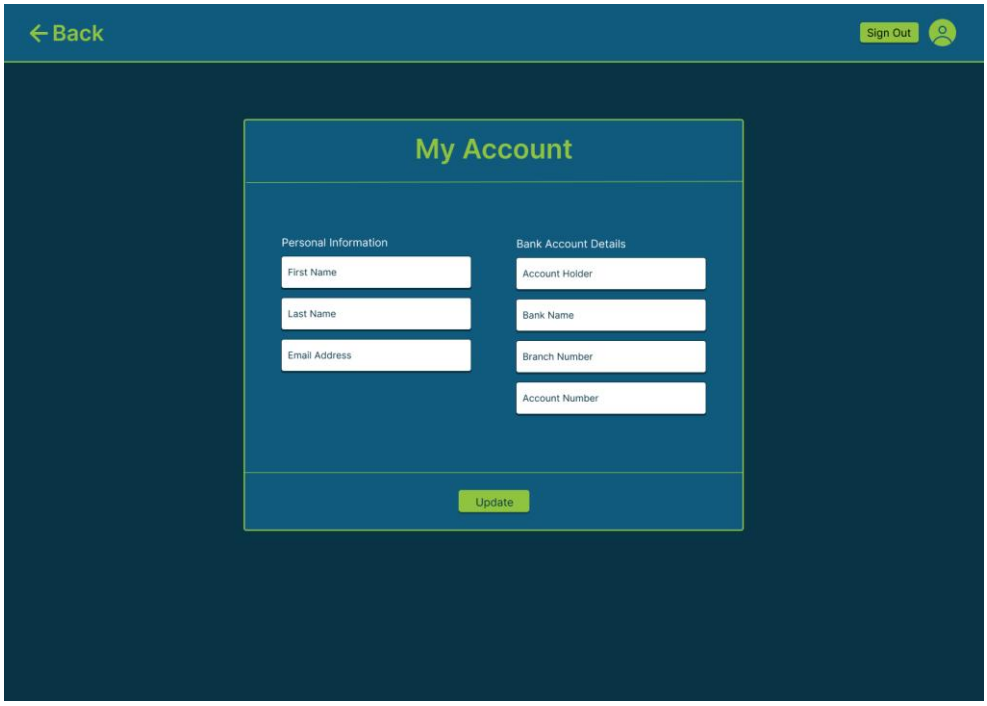


Figure 10 - User account view.

Project Plan

ID	Task Name	Start	End	Duration	Dependency
1	Contract Monthly Claim System	2025-09-11	2025-11-19	50 days	
2	Part 1	2025-09-11	2025-09-16	3.33 days	
3	Design Database	2025-09-11	2025-09-11	0.1 days	
4	Brainstorm Classes	2025-09-11	2025-09-11	0.02 days	
5	Create UML Class Diagram	2025-09-11	2025-09-11	0.08 days	4FS
6	Design GUI Prototype	2025-09-12	2025-09-16	2.33 days	
7	Select Colour Palette	2025-09-12	2025-09-12	0.2 days	
8	Design GUI in Figma	2025-09-12	2025-09-12	0.33 days	7FS
9	Build MVC UI Prototype	2025-09-16	2025-09-16	0.33 days	8FS+1.47 days
10	Create Documentation	2025-09-15	2025-09-16	1.22 days	
11	Explain Database Design	2025-09-15	2025-09-15	0.04 days	5FS+1.9 days
12	Explain GUI Design	2025-09-15	2025-09-15	0.08 days	8FS+0.47 days
13	Detail Assumptions & Constraints	2025-09-15	2025-09-15	0.04 days	11FS,12FS
15	Review Documentation	2025-09-16	2025-09-16	0.1 days	13FS+1 day
14	Documentation Completed	2025-09-16	2025-09-16	0 days	15FS
16	Part 2	2025-10-06	2025-10-13	5.58 days	
17	System Storage	2025-10-06	2025-10-06	0.37 days	
18	Create Database	2025-10-06	2025-10-06	0.33 days	14FS+13.78 days
19	Create FileShare for Supporting Docume…	2025-10-06	2025-10-06	0.04 days	18FS
20	Create FileShare for Invoices	2025-10-06	2025-10-06	0.04 days	18FS
21	Claim Submission	2025-10-07	2025-10-09	2.66 days	
22	Implement Claim CRUD	2025-10-07	2025-10-08	2 days	18SS+1 day
23	Implement Supporting Document Upload	2025-10-09	2025-10-09	0.33 days	22FS,19FS
24	Implement Error Handling	2025-10-09	2025-10-09	0.33 days	23FS
25	Claim Approval	2025-10-10	2025-10-10	0.58 days	
26	Implement Claim Status Update	2025-10-10	2025-10-10	0.25 days	22FS+1 day

Figure 11 - Project Plan Pt. 1

27	Implement Error Handling	2025-10-10	2025-10-10	0.33 days	26FS
28	▼ Unit Testing	2025-10-09	2025-10-13	2.25 days	
29	Claims CRUD	2025-10-13	2025-10-13	0.33 days	26FS+1 day
30	Document Upload	2025-10-09	2025-10-09	0.25 days	23FS
32	Core Features Completed	2025-10-13	2025-10-13	0 days	29FS
33	▼ Part 3	2025-11-14	2025-11-19	4 days	
34	▼ Claim Submission Automation	2025-11-14	2025-11-14	0.33 days	
35	Implement Input Validation	2025-11-14	2025-11-14	0.33 days	32FS+23.42 days
36	▼ Claim Verification Automation	2025-11-14	2025-11-17	1 day	
37	Implement Rule-based Verification	2025-11-14	2025-11-17	1 day	35FS
38	▼ HR Task Automation	2025-11-17	2025-11-18	1.67 days	
39	Implement Invoice Generation	2025-11-17	2025-11-18	1 day	37FS
40	Implement Report Generation	2025-11-18	2025-11-18	0.67 days	37FS+1 day
41	PowerPoint Presentation	2025-11-19	2025-11-19	1 day	40FS
42	CMCS Project Completed	2025-11-19	2025-11-19	0 days	41FS

Figure 12 - Project Plan Pt. 2

This project plan has been created with the BCAD2 programme assessment schedule in mind to remain achievable. The CMCS will be developed in parallel other projects, hence the large gaps between milestones.

Assumptions & Constraints

- The CMCS will be used by the PCs, AMs, and CLs at VC and should be designed with that in mind.
- Users' technical ability will be a broad range.
- The CMCS will be a ASP.NET MVC web app to be used on desktop or laptop computers, primarily.
- Storage will be both relational database and file storage.
- A single claim must be approved by one PC and one AM.
- A PC or AM is only authorised to approve claims made against modules which they have authority over.
- PC and AM accounts will be created and maintained by the system administrator.
- CL accounts will be created by the system administrator, but personal information and bank account details will be modifiable by the CL.
- For each CL, each month, on a specified date, a single invoice will be generated, the line items of which will be the LC's approved claims made that month.
- The CMCS itself will not process and payments.
- Information processing must comply with POPIA.
- Single Developer working on the project.
- Stakeholder engagement is limited to lecturer feedback after assignment submission.
- Prototype delivered by 17 September 2025.
- Core features delivered by 22 October 2025.
- Deadline is 21 November 2025.

(xAI, 2025).

References

xAI. 2025. Grok (Fast). [Large language model]. Available at: https://grok.com/share/c2hhcmQtNA%3D%3D_13829dde-1b56-484b-a1d9-afd9513e00c5 [Accessed: 15 September 2025].

Annexure A: Disclosure of AI Usage in my Assessment

Part 1

Assumptions & Constraints

Tool: Grok (Fast)

Purpose of use: to clarify/expand the question in order to provide a more informed answer.

Date(s): for the purpose of answering the question, I prompted Grok for help on 12 September 2025. However, I was not signed into Grok at the time and thus my chat history was not saved. On 15 September 2025, I signed into Grok and prompted the chatbot again, in order to provide a shareable link and screenshot for this AI usage declaration.

Link: https://grok.com/share/c2hhcmQtNA%3D%3D_13829dde-1b56-484b-a1d9-afd9513e00c5.

Screenshot:

